



Tutorial S1B: Probability

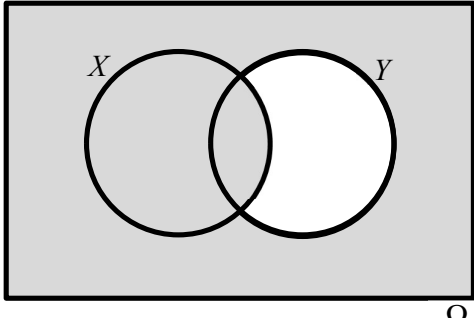
Section A (Basic Questions)

- 1 Two events X and Y and their respective probabilities of occurring are given by $P(X) = 0.6$ and $P(Y) = 0.7$. It is also given that $P(X \cup Y) = 0.95$. Calculate

- (a) $P(X \cap Y)$,
 (b) $P(Y|X)$,
 (c) $P(X \cup Y')$.

Determine if the events X and Y are independent.

[(a) 0.35 (b) $\frac{7}{12}$ (c) 0.65]

9233/J1983/02/Q3 (modified)	
Solution:	
(a)	$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$ $0.95 = 0.6 + 0.7 - P(X \cap Y)$ $P(X \cap Y) = 0.35$
(b)	$P(Y X) = \frac{P(Y \cap X)}{P(X)} = \frac{0.35}{0.6} = \frac{7}{12}$
(c)	 From the Venn diagram, $P(X \cup Y') = 1 - (P(Y) - P(X \cap Y))$ $= 1 - 0.7 + 0.35 = 0.65$
	$P(X \cap Y) = 0.35$ and $P(X) \times P(Y) = 0.6 \times 0.7 = 0.42$ Since $P(X \cap Y) \neq P(X) \times P(Y)$, events X and Y are not independent. OR Since $P(Y X) = \frac{7}{12} \neq P(Y) = 0.7$, events X and Y are not independent.

- 2 A die is known to be biased in such a way that, when it is thrown, the probability of a 6 showing is $\frac{1}{4}$. This biased die and an ordinary fair dice are thrown. Find the probability that
- (a) the fair die shows a 6 and the biased die does not show a 6,
 - (b) at least one of the two dice shows a 6,
 - (c) exactly one of the two dice shows a 6, given that at least one of them shows a 6.
- [(a) $\frac{1}{8}$ (b) $\frac{3}{8}$ (c) $\frac{8}{9}$]

9233/1998/02/Q3

Solution:

(a)	P(fair die shows a 6 and the biased die does not show a 6) $= \frac{1}{6} \times \frac{3}{4} = \frac{1}{8}$
(b)	P(at least one of the two dice shows a 6) $= 1 - P(\text{none of the dice shows a 6})$ $= 1 - \frac{5}{6} \times \frac{3}{4} = \frac{3}{8}$
(c)	P(exactly one of the two dice shows a 6 at least one of the two dice shows a 6) $= \frac{P(\text{exactly one of the two dice shows a 6 and at least one of the two dice shows a 6})}{P(\text{at least one of the two dice shows a 6})}$ $= \frac{P(\text{exactly one of the two dice shows a 6})}{P(\text{at least one of the two dice shows a 6})}$ $= \frac{\frac{1}{6} \times \frac{3}{4} + \frac{5}{6} \times \frac{1}{4}}{\frac{3}{8}}$ $= \frac{8}{9}$

- 3 In the first stage of a computer game, the player chooses, at random, one of 5 icons, only one of which is correct. If the correct icon is chosen then, in the second stage, the player chooses, at random, one of 8 icons, only one of which is correct. If an incorrect icon is chosen in the first stage then, in the second stage, the player chooses, at random, one of 10 icons; only one of which is correct.

The events A and B are defined as follows:

A : the first icon chosen is correct.

B : the second icon chosen is correct.

Find

(i) $P(A \cap B)$, [1]

(ii) $P(B)$, [3]

(iii) $P(A \cup B)$, [2]

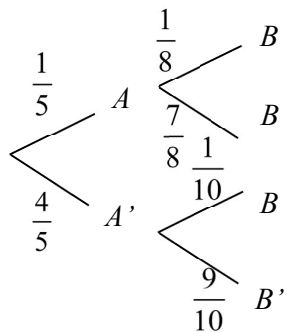
(iv) $P(A|B)$. [2]

[(i) $\frac{1}{40}$ (ii) $\frac{21}{200}$ (iii) $\frac{7}{25}$ (iv) $\frac{5}{21}$]

9233/2003/02/Q25

Solution:

(i) $P(A \cap B) = \left(\frac{1}{5}\right)\left(\frac{1}{8}\right) = \frac{1}{40}$



(ii) $P(B) = P(A \cap B) + P(A' \cap B)$
 $= \frac{1}{40} + \left(\frac{4}{5}\right)\left(\frac{1}{10}\right) = \frac{21}{200}$

(iii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= \frac{1}{5} + \frac{21}{200} - \frac{1}{40} = \frac{7}{25}$

(iv) $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{40}}{\frac{21}{200}} = \frac{5}{21}$

- 4 A bag contains 3 red balls and 3 green balls. Balls are drawn from the bag at random, one by one and without replacement.

- (a) Show that the probability that the first 3 balls drawn are red is $\frac{1}{20}$.
- (b) Find the probability that the first 3 balls drawn consist of 2 red balls and 1 green ball (in any order). Hence, or otherwise show that the probability that the third red ball appears on the fourth draw is $\frac{3}{20}$.

[(b) $\frac{9}{20}$]

9233/1995/02/Q6 (modified)

Solution:

- (a) P(first 3 balls drawn are red)

$$= \frac{3}{6} \times \frac{2}{5} \times \frac{1}{4} \quad \text{OR} \quad \frac{{}^3C_3}{{}^6C_3}$$

$$= \frac{1}{20}$$

- (b) P(first 3 balls drawn consist of 2 red balls and 1 green ball in any order)

$$= P(RRG) + P(RGR) + P(GRR)$$

$$= \frac{3}{6} \times \frac{2}{5} \times \frac{3}{4} + \frac{3}{6} \times \frac{3}{5} \times \frac{2}{4} + \frac{3}{6} \times \frac{3}{5} \times \frac{2}{4} \quad \text{OR} \quad \frac{{}^3C_2 \times {}^3C_1}{{}^6C_3}$$

$$= \frac{9}{20}$$

OR $\frac{3}{6} \times \frac{2}{5} \times \frac{3}{4} \times \frac{3!}{2!}$

OR $\frac{3}{6} \times \frac{2}{5} \times \frac{3}{4} \times {}^3C_2$

P(third red ball appears on the fourth draw)

= P(first 3 balls drawn consist of exactly 2 red balls **and** third red ball appears on fourth draw)

$$= \frac{9}{20} \times \frac{1}{3} = \frac{3}{20}$$